

# Acute Vision Loss

Andrew Kuk  
Clinical Assistant Professor

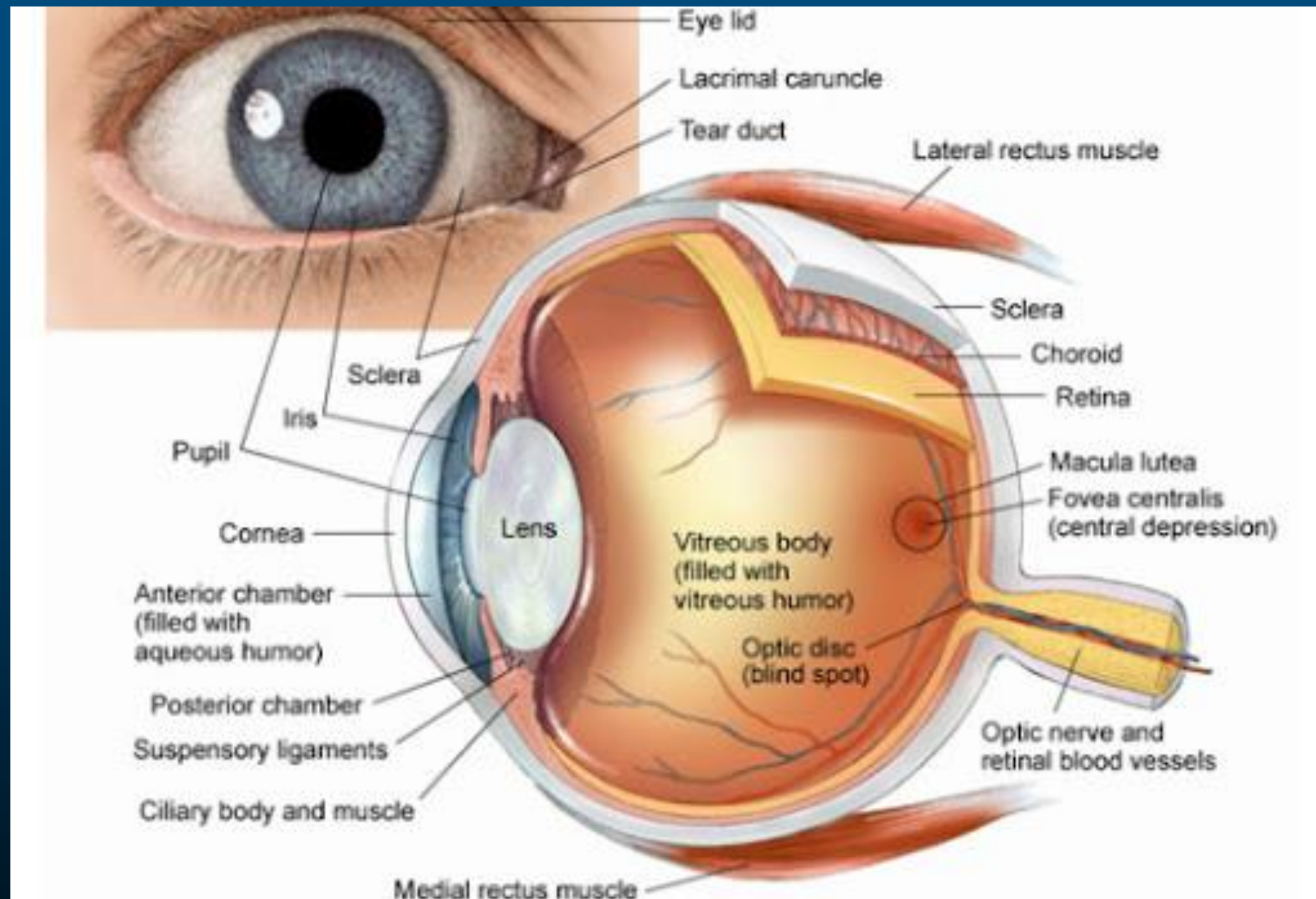
(Slides adapted from Dr Nicholas Fung's)



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# What happens during acute vision loss?



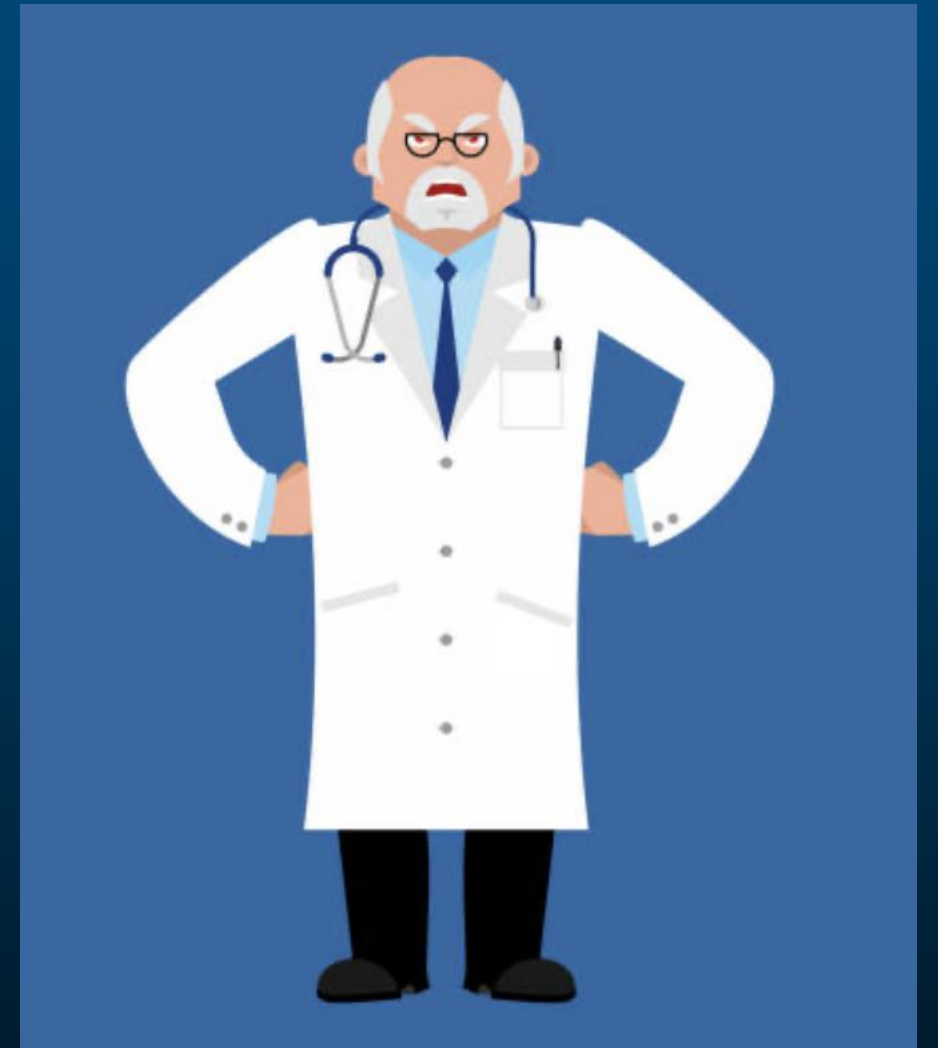
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# What are the causes?

Remember to categorize your answers!

- 1) Transient vs Persistent
- 2) Painful vs Painless
- 3) Anatomically (start anterior to posterior)



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# So what do we do?

## History:

P/C: Onset, duration, pain, trauma, foreign body, contact lens user

POHx: Previous eye surgeries / lasers, previous episodes

PMHx: Cardio (HTN / lipid /DM), neuro (CVA / TIA / demyelinating or AI disease), haem (bleeding tendencies)

## Examination:

Visual Acuity, IOP, anterior segment, posterior segment by dilated fundal exam

## Investigations:

Anterior segment imaging: Specular microscopy, Optical Coherence Tomography (OCT), keratometry, slit lamp photo

Posterior segment imaging: ultrasound B scan, OCT macula, fluoresceine fundus angiography (FFA)

Systemic work up: Brain and orbit CT/MRI, Bloods, ERG / EOG

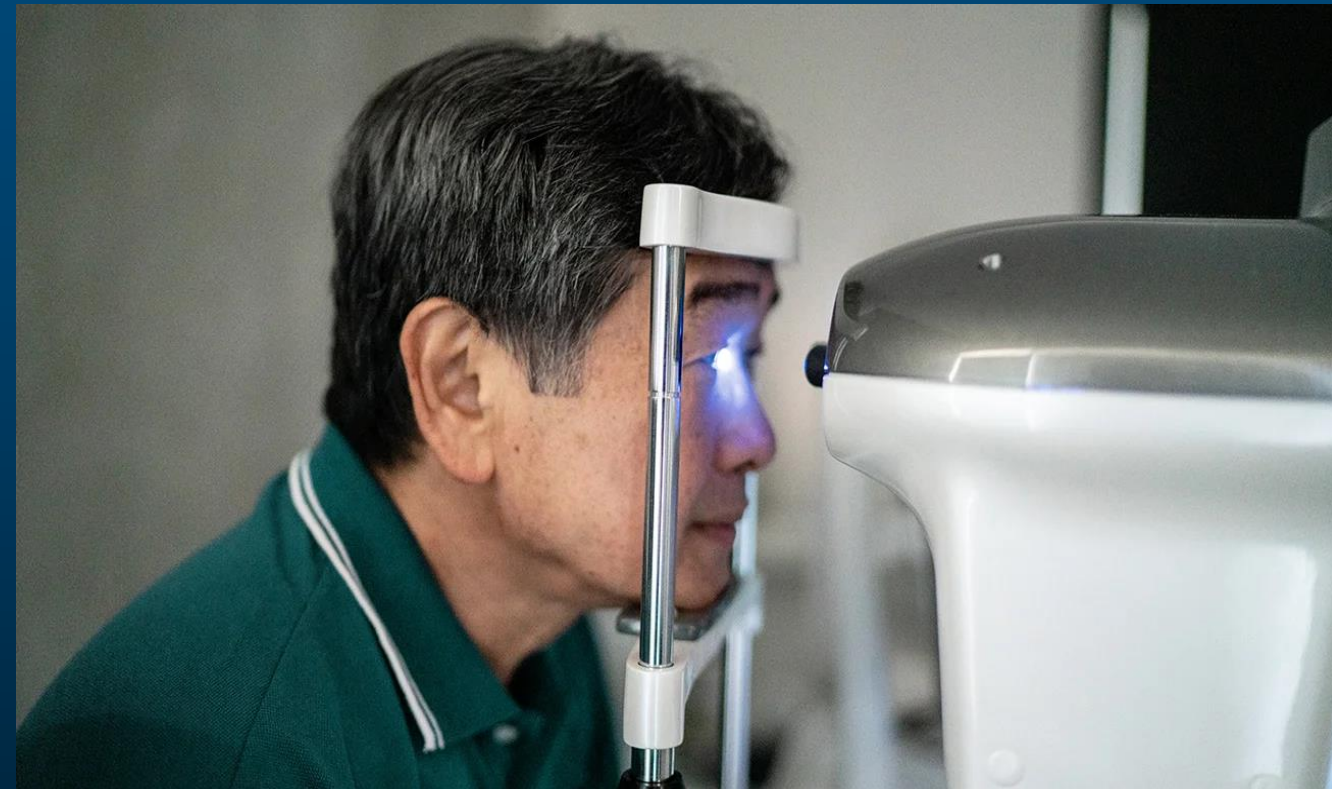
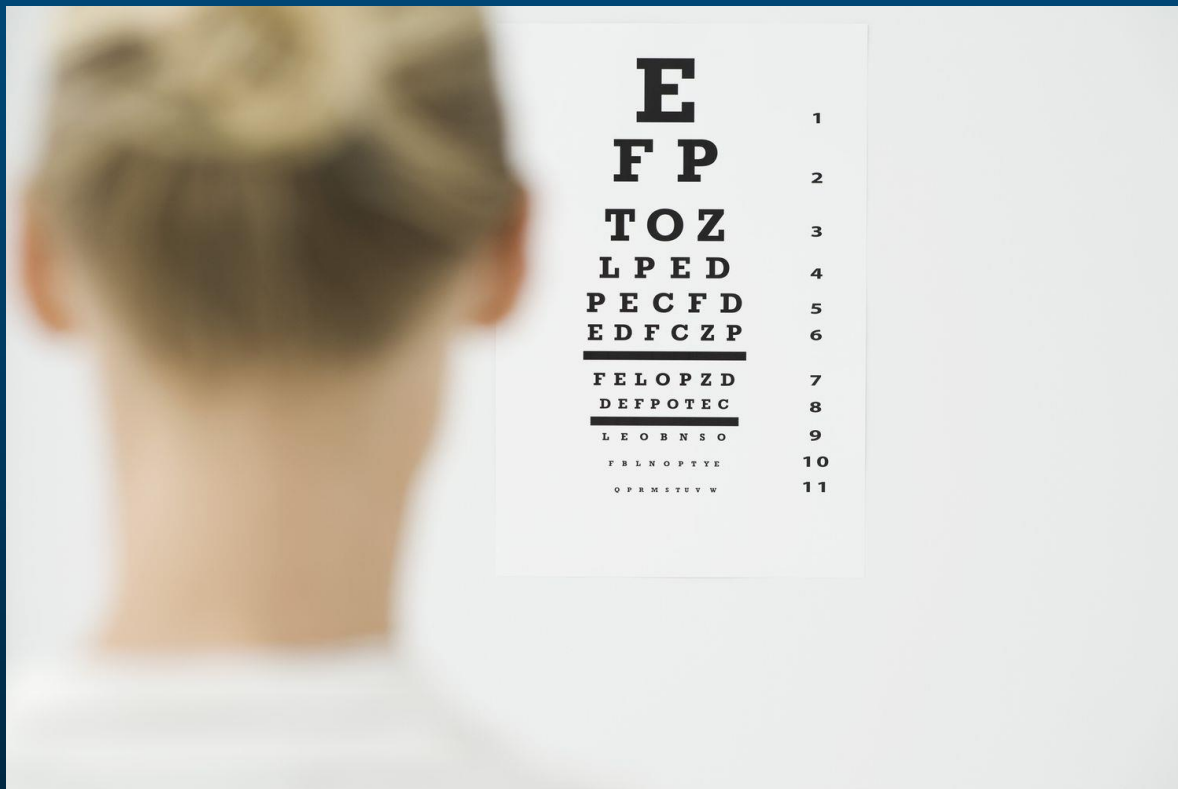


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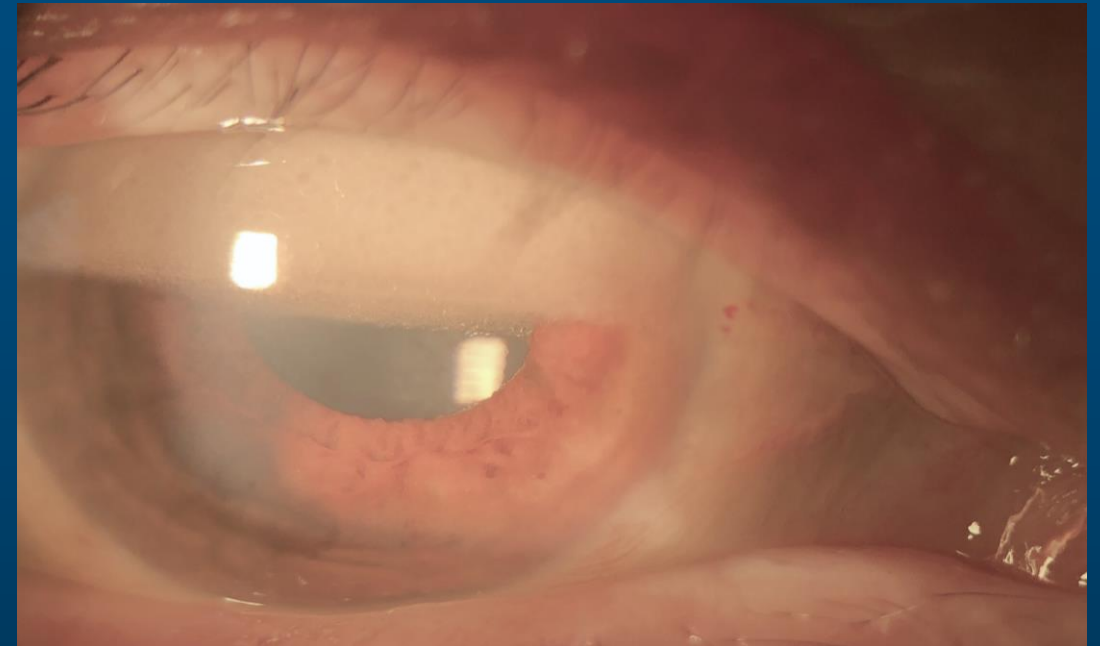
# Examination (front to back!)



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# Examination: Slit lamp of anterior segment



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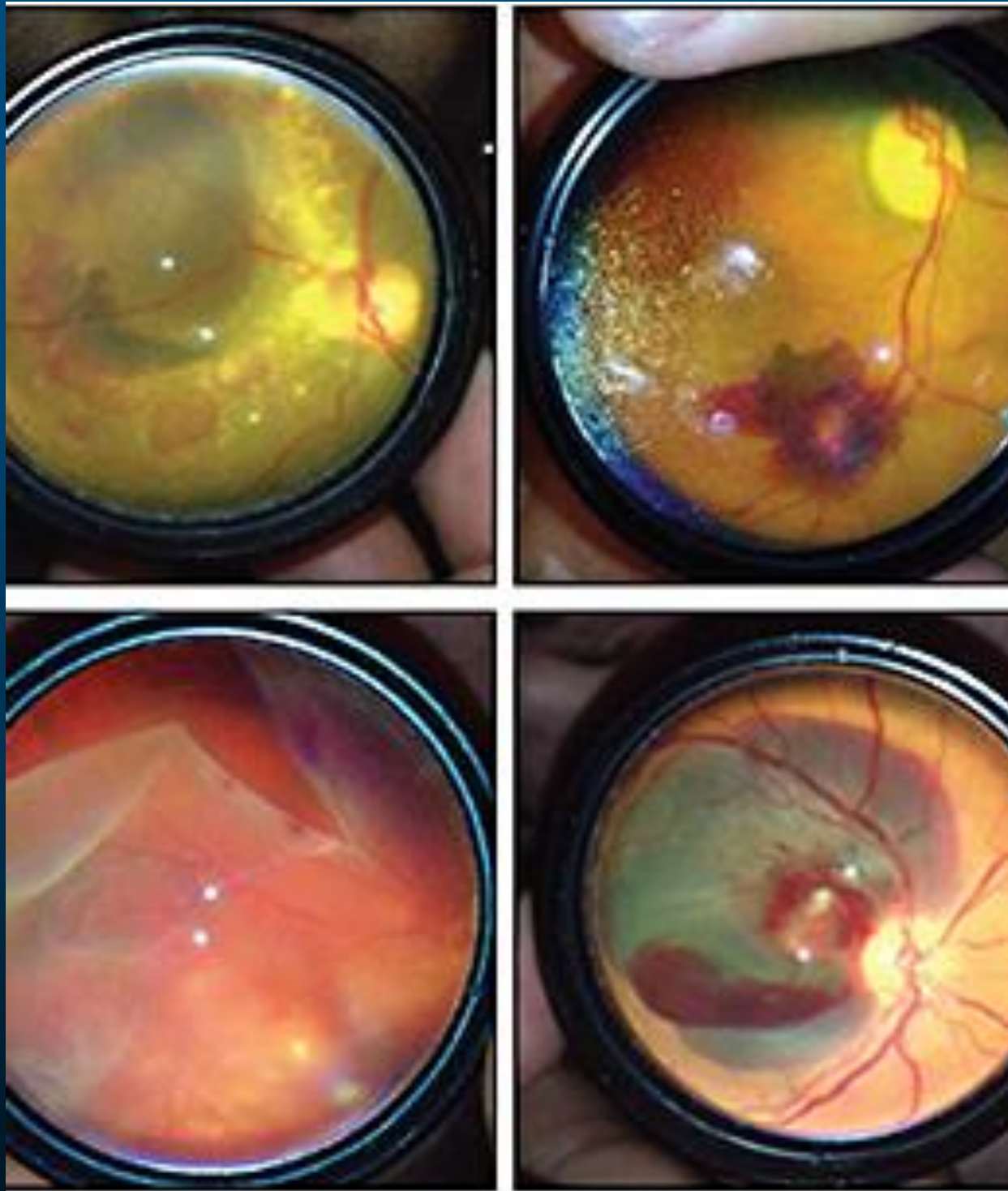
# Examination: Slit lamp of posterior segment



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# Examination: Slit lamp of posterior segment



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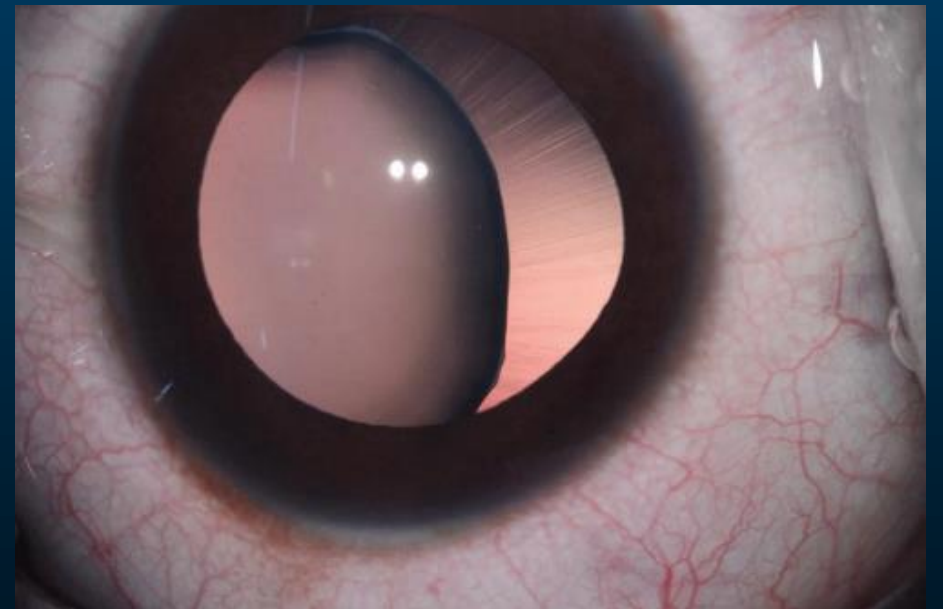
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# What are the causes?

Transient (also keep front to back!)

- 1) Glare
- 2) Cornea surface disruption
- 3) Lens subluxation (not dislocated yet!)
- 4) Amaurosis fugax (TIA)
- 5) Migraine (vasospasm)



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# What are the causes?

Persistent (also keep front to back!)

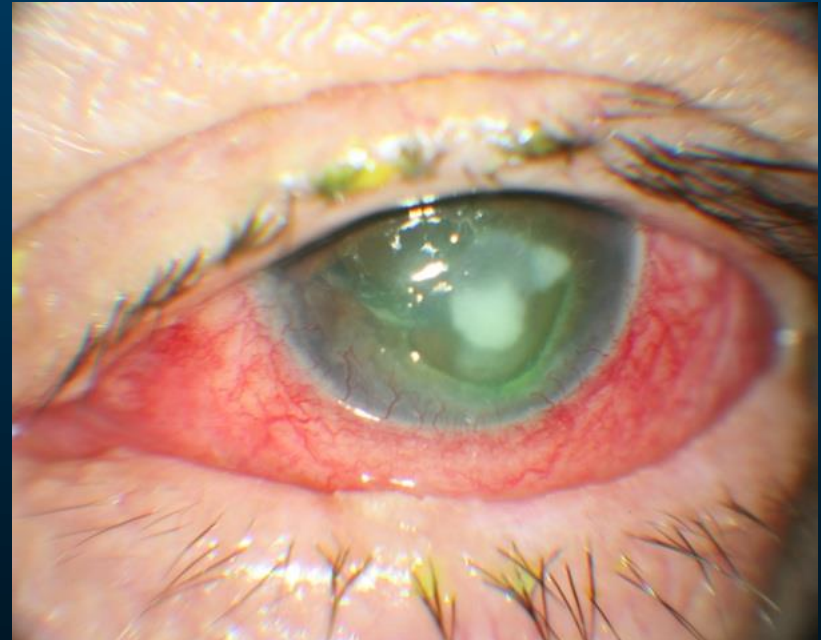
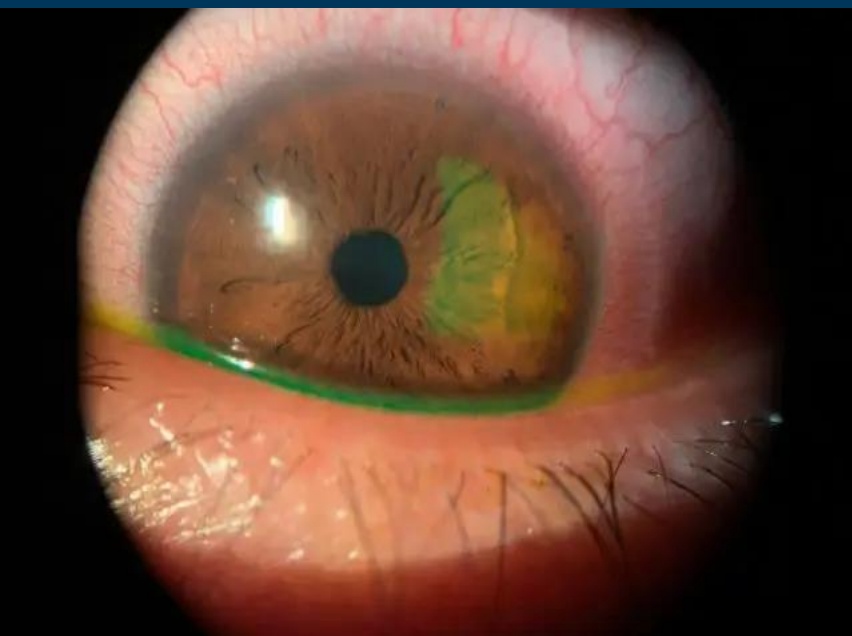
Cornea:

Epithelial defect !

Foreign body !

Oedema (due to inflamm / high IOP)

Keratitis !



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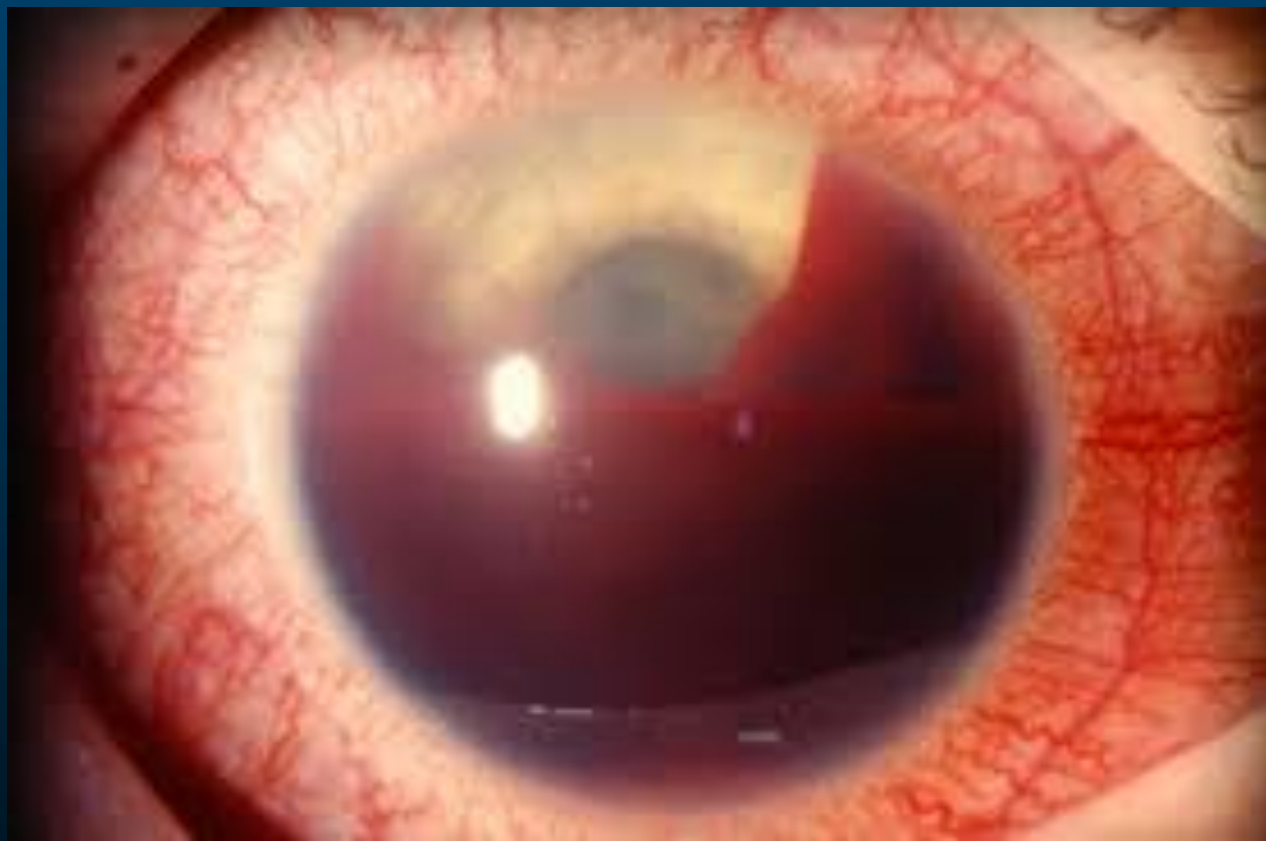
# What are the causes?

Persistent (also keep front to back!)

Anterior Chamber:

Hyphaema (trauma ! vs bleeding disorder)

Uveitis +/- ! / hypopyon (idio, reactive, infective, AI)



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# What are the causes?

Persistent (also keep front to back!)

Anterior Chamber:

Acute angle closure glaucoma !



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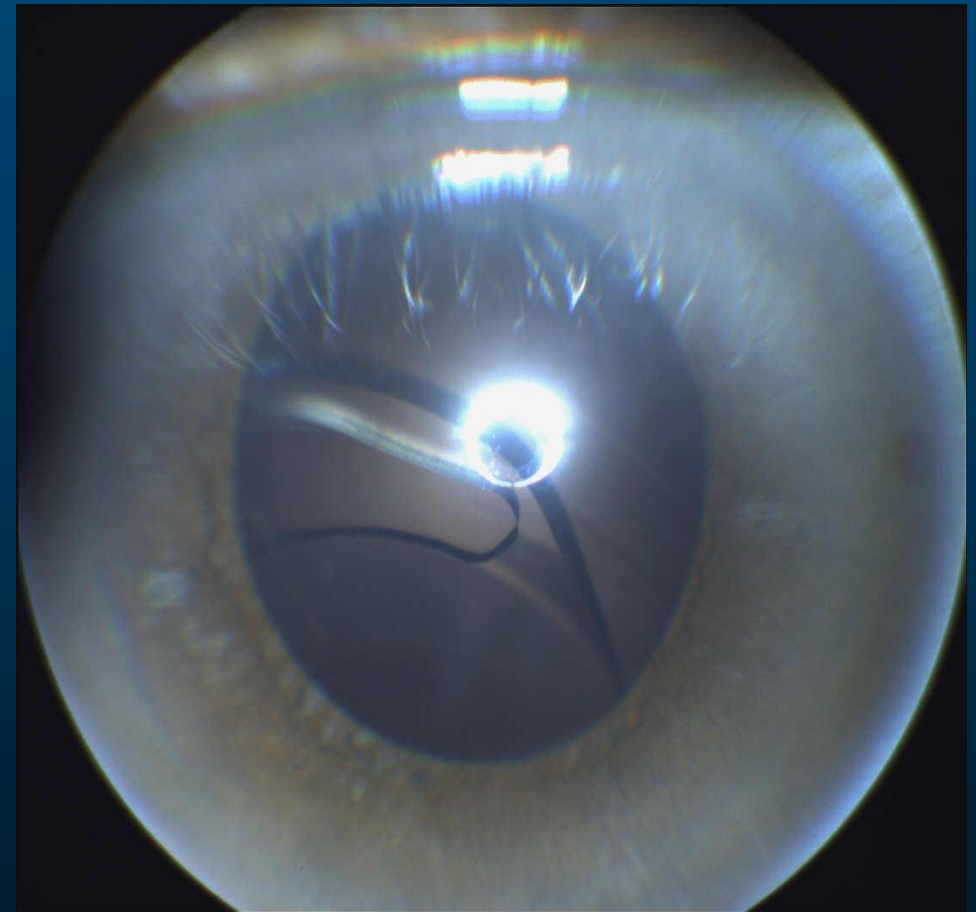
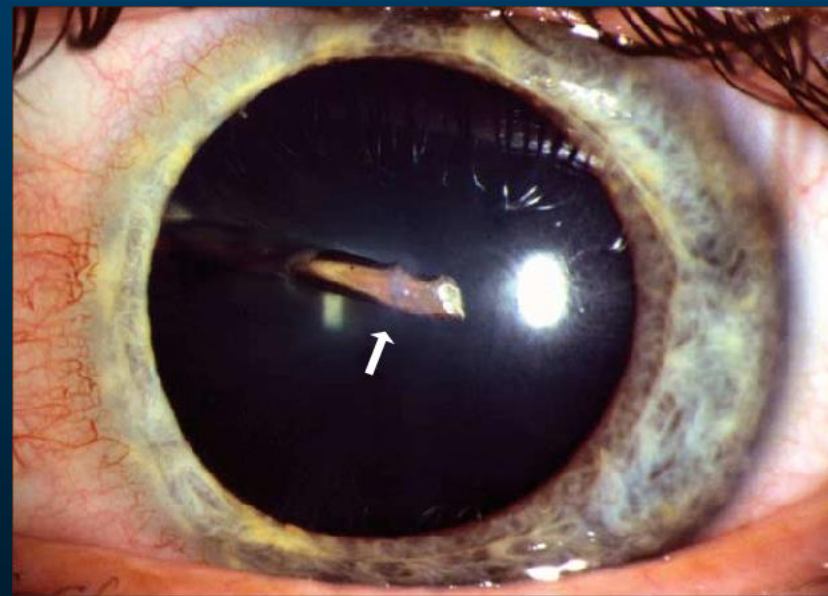
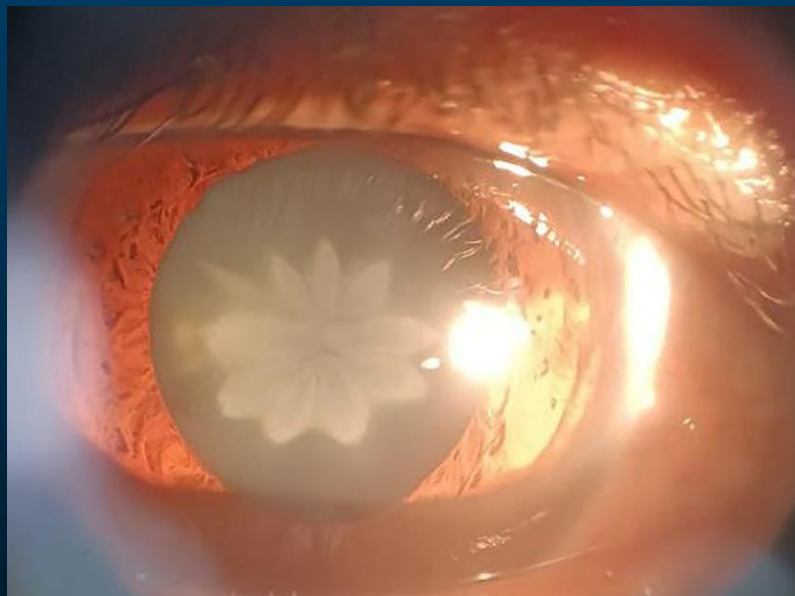
# What are the causes?

Persistent (also keep front to back!)

Lens:

Post traumatic cataract / foreign body

Dislocation



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# What are the causes?

Persistent (also keep front to back!)

## Vitreous:

Haemorrhage

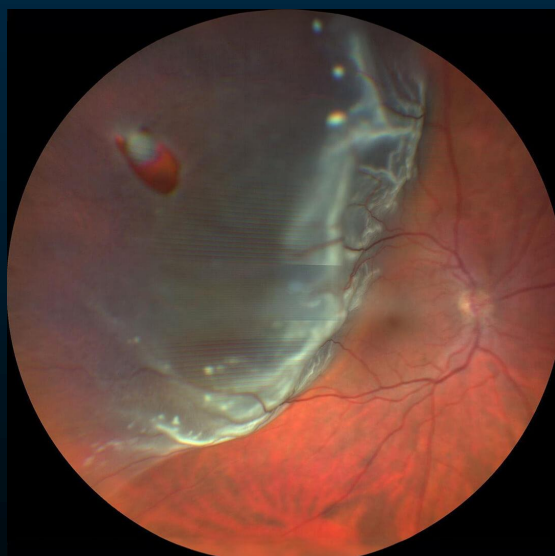
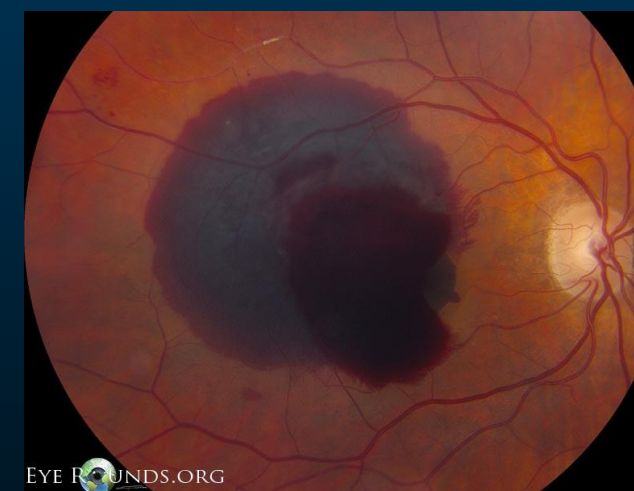
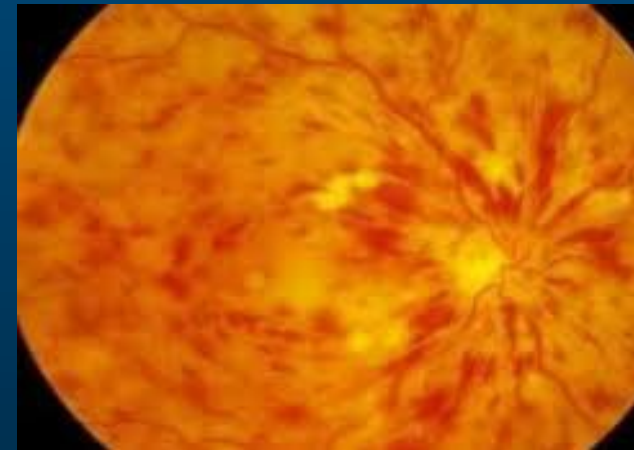
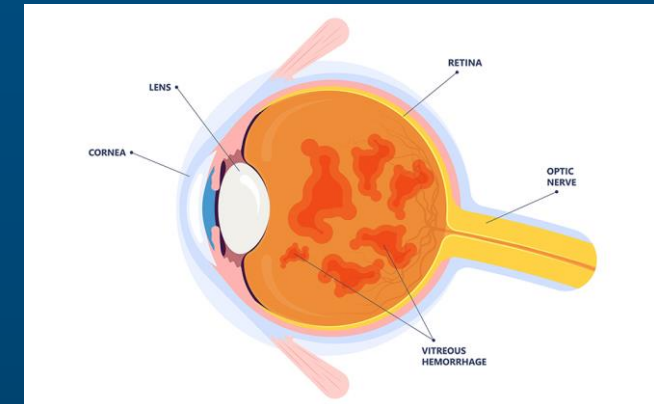
Vitritis

## Retina:

Vascular occlusions (CRVO / BRVO and CRAO / BRAO)

Haemorrhage or oedema at macula

Detachment



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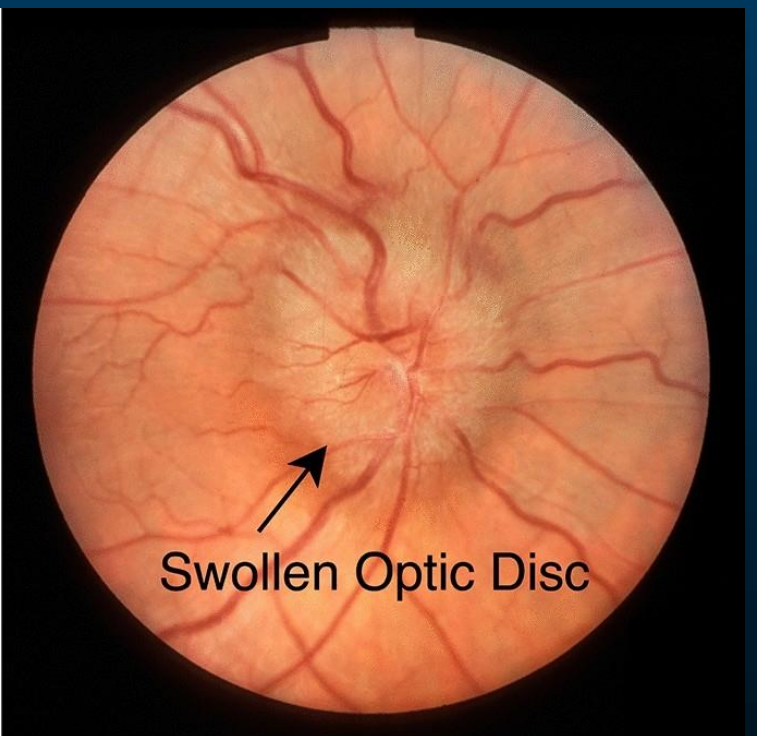
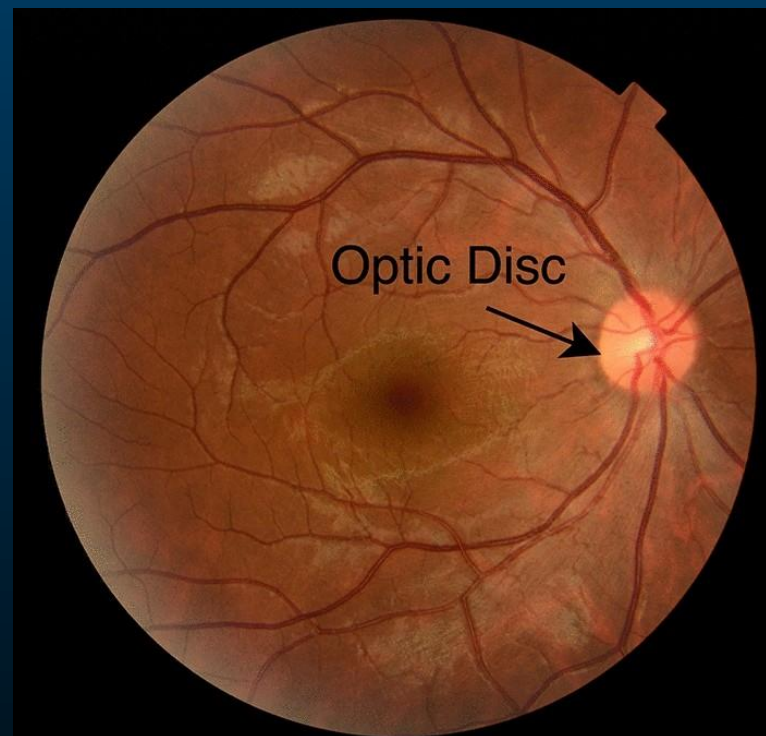
# What are the causes?

Persistent (also keep front to back!)

Optic Disc:

Swelling = optic neuritis ! vs AAION (GCA) ! or NAION (ischaemia) vs papilloedema

Avulsion



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# What are the causes?

## Transient (front to back)

1) Glare

2) Cornea surface disruption

3) Lens subluxation

4) Amaurosis fugax (TIA)

5) Migraine (vasospasm)

## Persistent (front to back)

### Cornea:

Epithelial defect, Foreign body  
Oedema  
Keratitis

### Anterior Chamber:

Hyphaema  
Uveitis  
Acute angle closure glaucoma

### Lens:

Post traumatic cataract / foreign body  
Dislocation

### Vitreous:

Haemorrhage  
Vitritis

### Retina:

Vascular occlusions (CRVO / BRVO and CRAO / BRAO)  
Haemorrhage or oedema of macula  
Detachment

### Optic disc:

Swelling (optic neuritis / ischaemic neuropathy / papilloedema)  
Avulsion



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# What are the causes?

## Painless (front to back)

### External:

Glare

### Cornea:

Cornea surface disruption

### Anterior chamber:

Hyphaema (bleeding disorder)

### Lens

Lens subluxation / dislocation (non traumatic)

### Vitreous:

Vitreous hemorrhage

Vitritis

### Retina:

Retinal vascular occlusions

Macula haemorrhage or oedema

Retinal detachment

### Other:

Amaurosis fugax (TIA)

Migraine (vasospasm)

## Painful (front to back)

### Cornea:

Epithelial defect

Foreign body

Oedema

Keratitis

### Anterior Chamber:

Hyphaema (trauma)

Uveitis (idio, reactive, infective, AI)

Acute angle closure glaucoma

### Lens:

Post traumatic cataract / foreign body

Traumatic subluxation / Dislocation

### Optic disc:

Swelling (optic neuritis / ischaemic neuropathy / papilloedema)

Avulsion



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# What are the causes?

## Anatomical (front to back)

### External:

Glare

### Cornea:

Cornea surface disruption

Epithelial defect

Foreign body

Oedema

Keratitis

### Anterior Chamber:

Hyphaema

Uveitis

Acute angle closure glaucoma

### Lens:

Post traumatic cataract / foreign body

Traumatic subluxation / Dislocation

### Vitreous:

Vitreous hemorrhage

Vitritis

### Retina:

Retinal vascular occlusions

Macula haemorrhage or oedema

Retinal detachment

### Optic disc:

Swelling (optic neuritis / ischaemic neuropathy / papilloedema)

Avulsion

### Other:

Amaurosis fugax (TIA)

Migraine (vasospasm)



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# Deep dive on some of the important causes

1. Keratitis
2. Acute angle closure glaucoma
3. Uveitis
4. Vascular occlusions
5. Vitreous hemorrhage
6. Macula hemorrhage / oedema
7. Retinal detachment
8. Optic disc swelling
9. \*Endophthalmitis\*



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# 1. Keratitis

## History:

Coryzal illness (viral)

Trauma (unusual organisms like fungal)

Contact lens user (bacterial, amoeba)

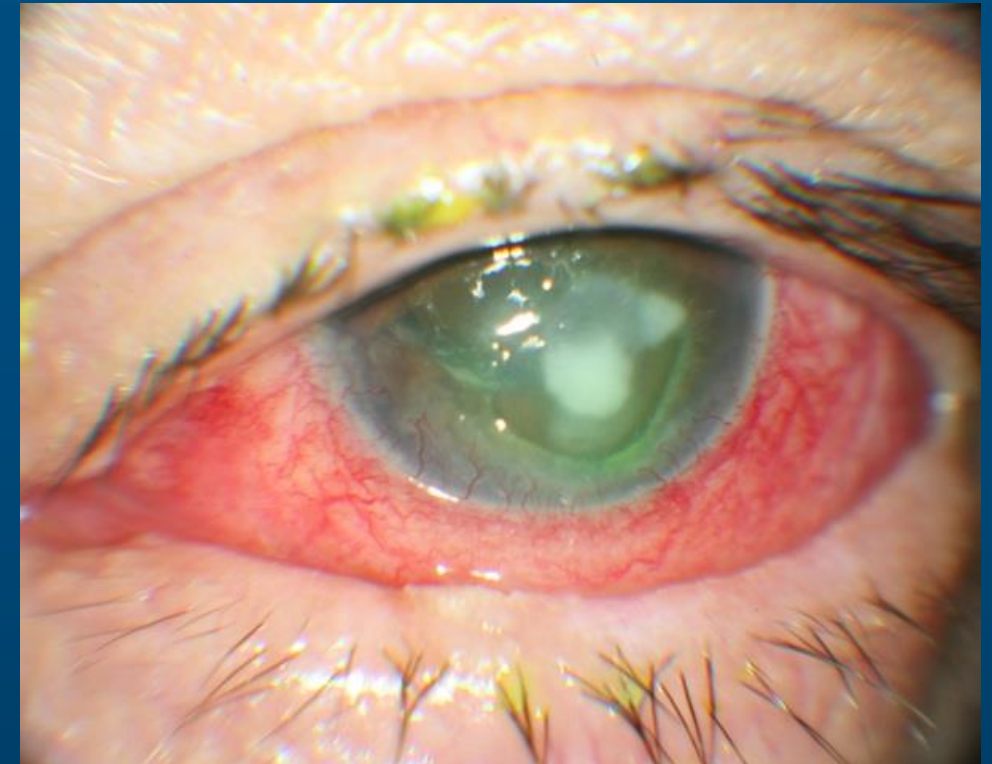
Dirty water exposure (amoeba)

## Management:

Recognise early!

Initiate topical treatment early

Refer to ophth for cornea scraping and adjustment of Rx



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# 2. Acute angle closure glaucoma

## History:

Preceding event such as reading in dim light

On cold Rx (antihistamines used as decongestant lead to mild pupil dilation + unknown narrow angle = AACG)

## Acute Symptoms:

Blurring, eye pain, headache, nausea

## Acute Signs:

Injected (redness), mid dilated pupils, shallow AC, corneal oedema, decreased visual acuity, raised IOP

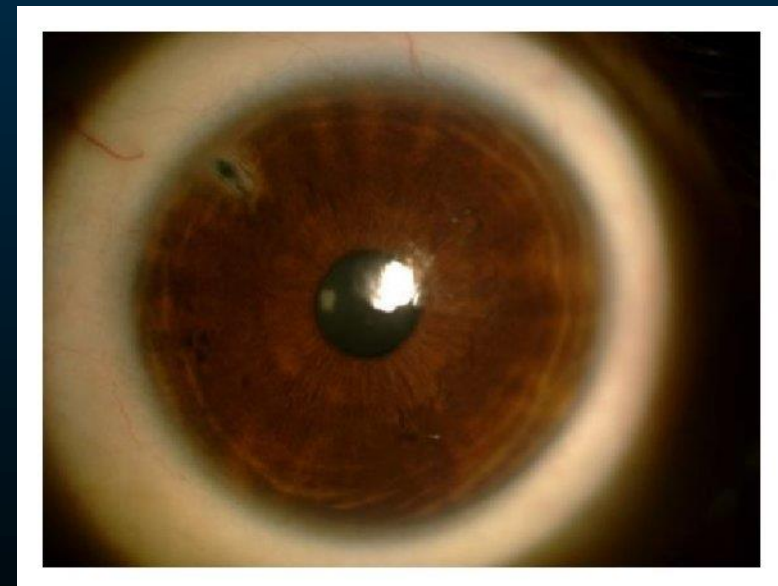
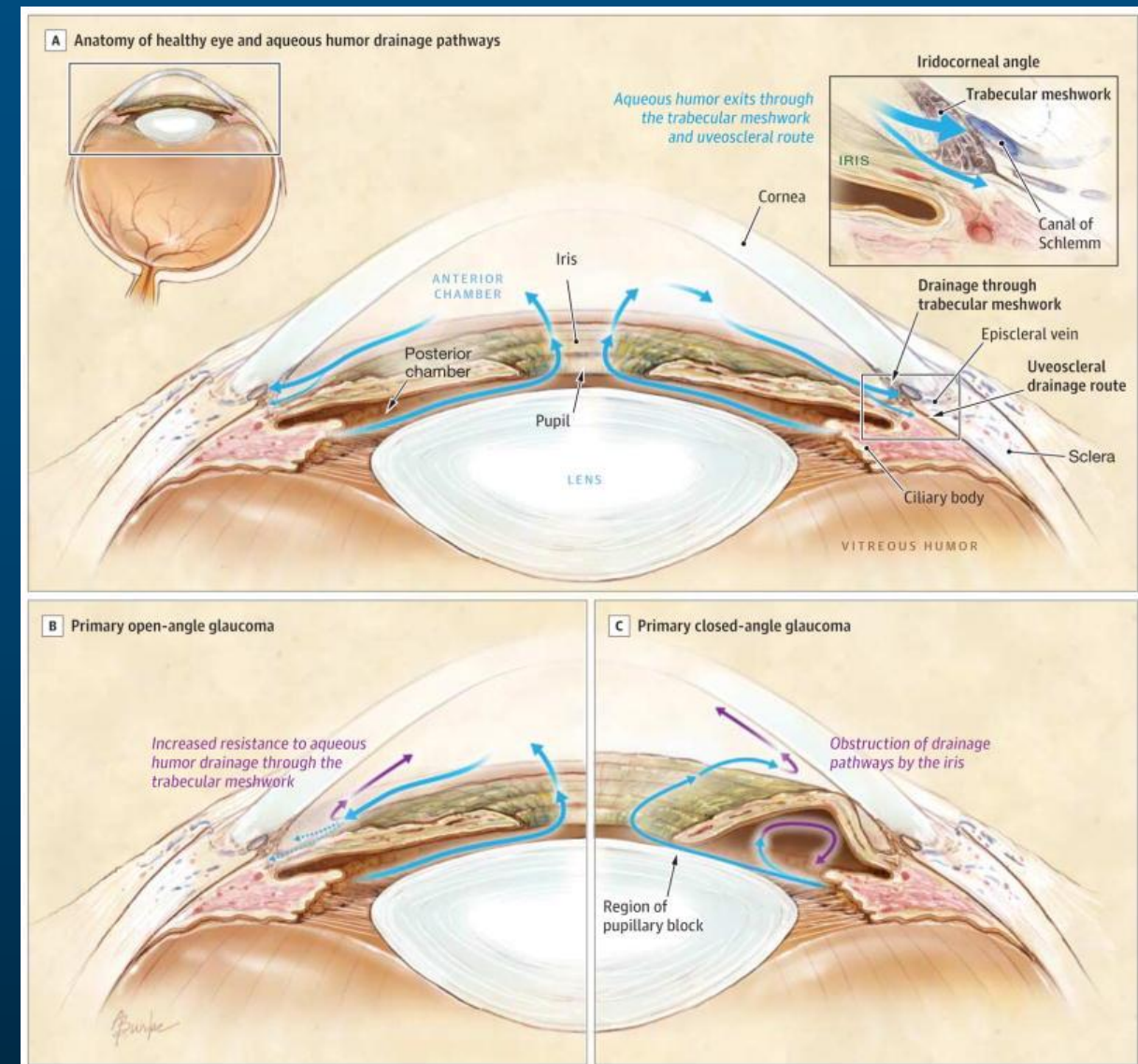
## Management:

Recognise early and consult ophth!

Constrict pupil - break the cycle with pilocarpine

Lower IOP: ALL topical agents and systemic agents

Ophth: laser ALPI / PI, surgical PI, early cataract OT



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# 3. Uveitis

## History:

Trauma , eye surgery, previous episodes, AI disease

## Ophthalmic signs:

Anterior Chamber cells, fibrin, flare, hypopyon, peripheral anterior synechiae (PAS) , posterior synechiae (PS), iris bombe, keratitic precipitates, corneal edema

## Causes:

Idiopathic (50%)

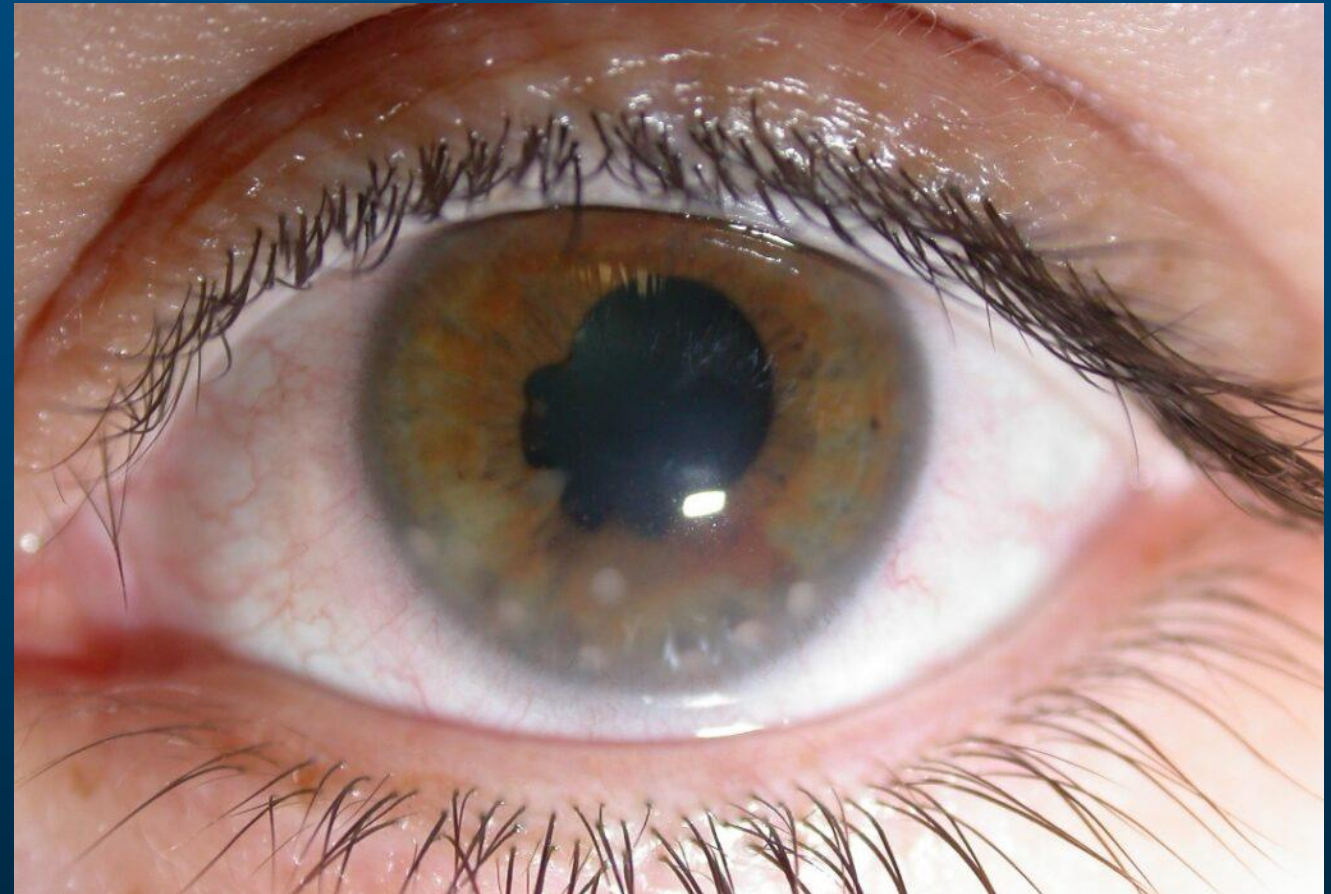
HLA B27 related (20%): Ankylosing Spondylitis, Reiter's disease, Inflammatory bowel disease, psoriasis

Viral (10%): herpes simplex, herpes zoster, CMV

Iatrogenic: surgical complications, trauma, implants (IOL), corneal transplant rejection

## Management:

Refer to ophth for pupil dilation (check no extension posteriorly or other causes in posterior manifesting as anterior uveitis), steroids

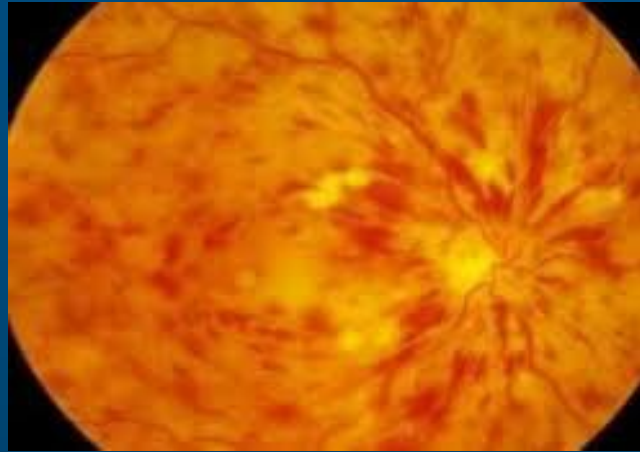


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# 4. Vascular occlusions



## RVO

Risk factor: Hypertension, diabetes, smoking, obesity, hyper coagulation disorders (Protein C and S deficiency, antiphospholipid syndrome), glaucoma

Systemic associations: hyperlipidemia, SLE, sarcoidosis, TB, syphilis, multiple myeloma, cryoglobulinemia, leukemia, lymphoma, sickle cell disease

### Classifications:

Branch RVO / Hemi RVO

Central RVO



## RAO

Risk factors: Hypertension, hyperlipidemia, hyperviscosity syndromes

Retina blood supplied by the retinal artery as it branches out from the optic disc

### Classifications:

Branch RAO

Central RAO (more common) retinal artery emboli



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# 5. Vitreous hemorrhage

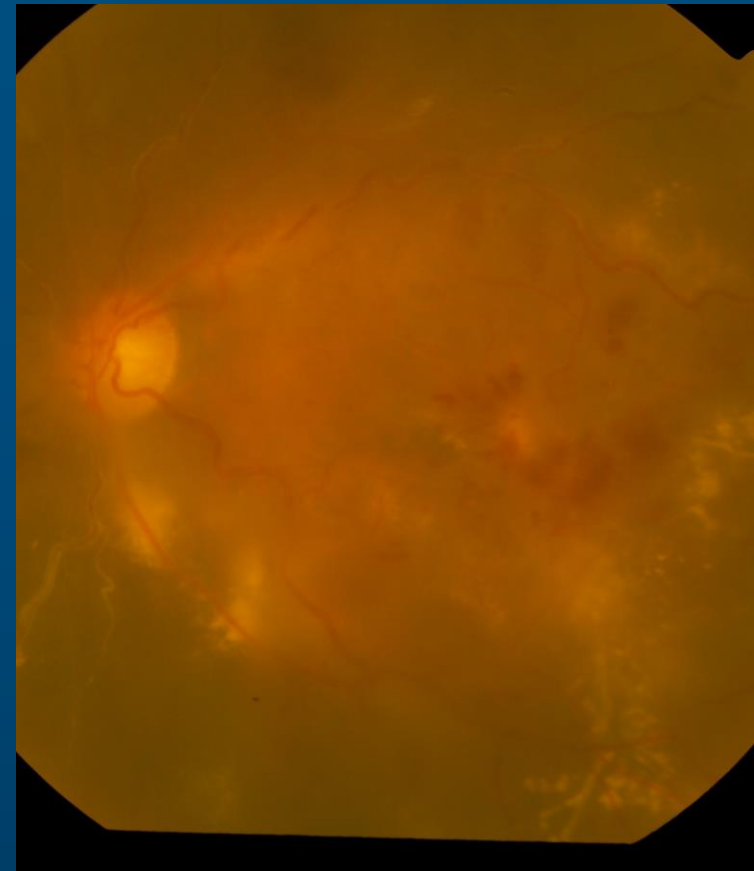
## Causes

Trauma

Neovascularization from:  
proliferative diabetic  
retinopathy, vascular occlusions

Posterior vitreous detachment /  
retinal tear

Bleeding tendencies



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## 6. Macula hemorrhage / oedema

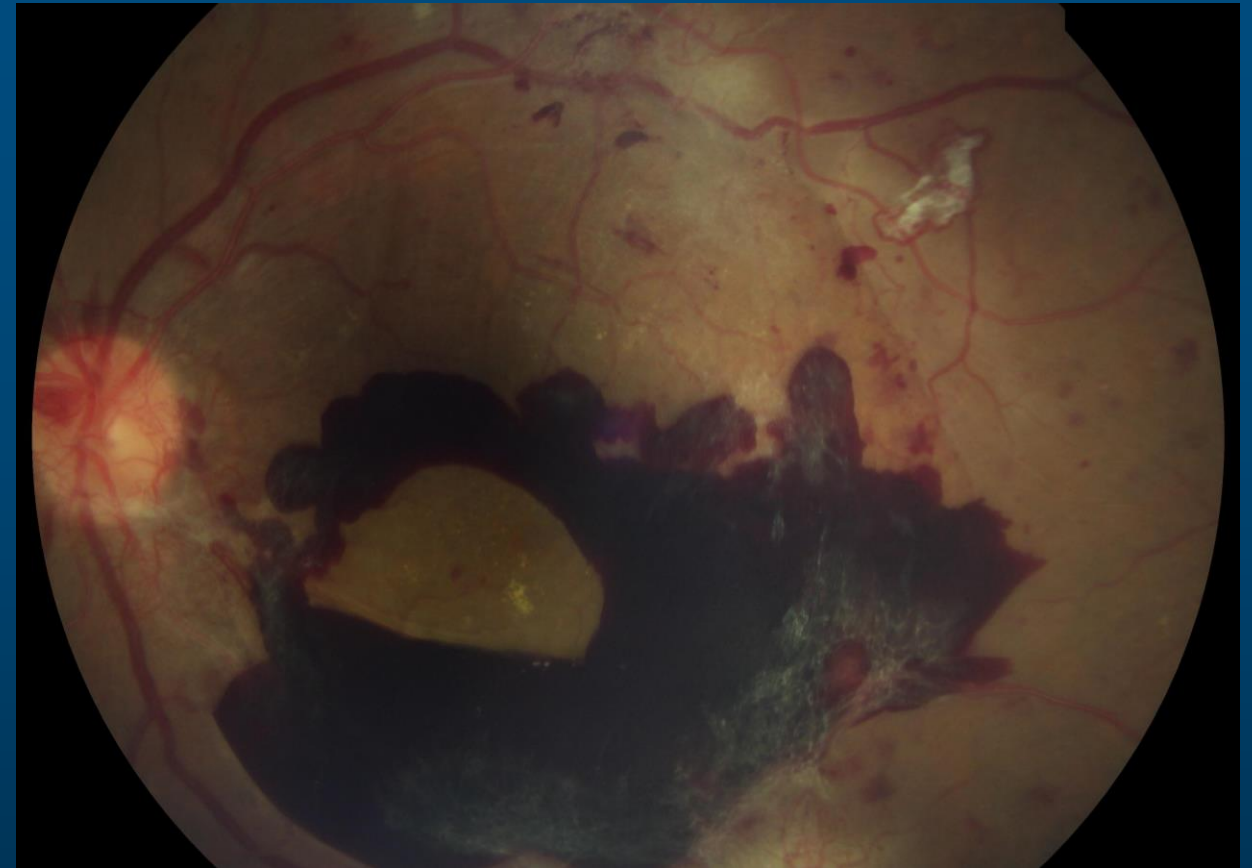


Dry AMD Mx:

Education , stop smoking, intake macular carotenoids (spinach, cabbage, broccoli), vitamins: AREDS2 supplement (Vit C, E, alpha carotene, and zinc), regular AMSLER grid monitoring

Wet AMD Mx:

Refer to ophth for antiVEGF injections / laser



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# 7. Retinal detachment

## Risk factors:

High myopia ( $>-6D$ ), family history, eye rubbing (eczema), systemic disease (Marfan's), trauma

## Presentation:

Persistent floaters / flashes

Progressively worsening visual field defect

## Management:

Recognise early and refer to ophth: laser / cryopexy / surgery



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# 8. Optic disc swelling

→ Demyelinating inflammation of the optic nerve



Normal optic disk

Edematous optic disk

## Clinical

- Sudden loss of monocular partial or complete vision
- Pain with movement of affected eye
- Afferent pupillary defect
- **Loss** of color (**red**) vision
- **Uhthoff's phenomenon**  
Transient worsening of vision with increased body temperature

## Etiologies

- **Multiple sclerosis** (most common)
- Infection (Lyme, Herpes, Syphilis)
- Autoimmune (Lupus, Neurosarcoidosis)
- Methanol poisoning
- B12 deficiency
- Diabetes

## Management

- Corticosteroids (intravenous)

|                       | ARTERITIC AION                                                 | NON ARTERITIC AION                                                              |
|-----------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------|
| MEAN AGE              | 70 years                                                       | 60 years                                                                        |
| GENDER                | Female > male                                                  | No relation                                                                     |
| ASSOCIATED SYMPTOMS   | Headache, scalp tenderness, jaw claudication                   | Occasional orbital pain                                                         |
| VISUAL ACUITY         | <6/60 in 76%                                                   | >6/60 in 61%                                                                    |
| OPTIC DISC APPEARANCE | Pale more than hyperemic edema<br>Normal to large cup          | Hyperemic more than pale edema<br>Small cup                                     |
| ESR                   | >70 (highly raised)                                            | 20-40 (mildly raised)                                                           |
| FFA                   | Choroidal (> 30 – 69 s) and disc filling delay                 | Disc filling delay                                                              |
| NATURAL HISTORY       | Poor prognosis for recovery<br>Fellow eye involved in upto 95% | Upto 3 line improvement in about 43% cases<br>Fellow eye involved in <30% cases |
| TREATMENT             | Urgent administration of corticosteroids                       | Doubtful role of corticosteroids                                                |

## Cause of Papilloedema

Idiopathic intracranial hypertension (IIH)

Venous sinus thrombosis

Hydrocephalus

Intracranial lesion (tumour, arterio-venous malformation, haemorrhage, aneurysm)



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# 9. Endophthalmitis

Not included into original list as not quite “acute” but is most destructive

Categorized into exogenous and endogenous

History: Recent ocular surgery, trauma, immunosuppressed / immunocompromised

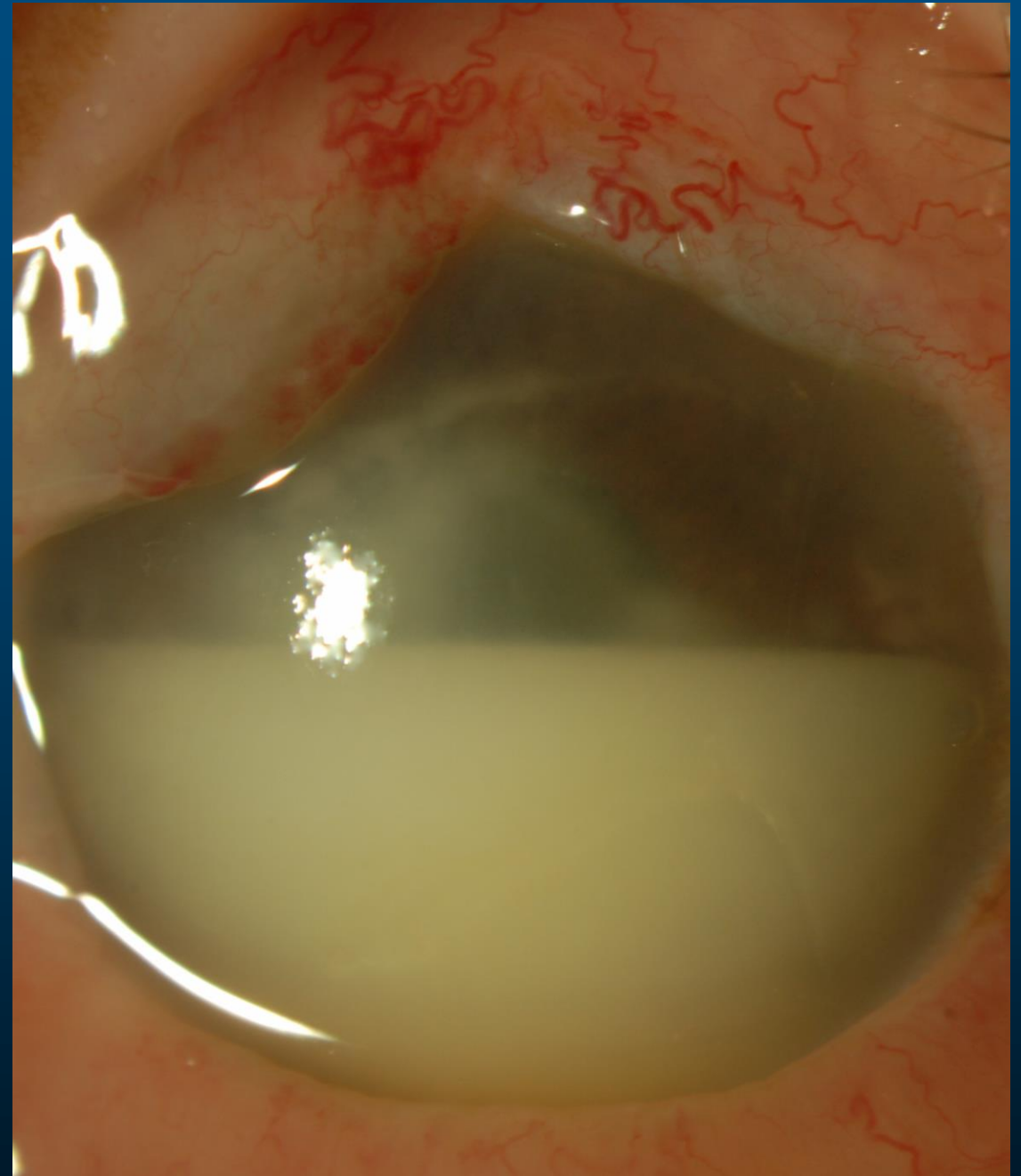
Recognize early! Low threshold of suspicion and consult ophth:

B scan, CT abdomen, U/S liver

Systemic antibiotics

Vitreous tap and inject antibiotics

Vitrectomy



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# SUMMARY

**Main questions:**

**Onset**

**Pain**

**Transient**

**Think of obvious reasons which the patient will likely be aware of:**

**Trauma**

**Contact lens**

**Recent eye surgery**

**Hint:**

**Painful: More anterior**

**Painless: More posterior**

**Transient: Systemic vascular issues**

**Persistent: Everything else! Go anatomical front to back**



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SC Whole Class Session

# TELL US WHAT YOU THINK



Look for today's teacher (Andrew Kuk)  
and any others you may have missed!

THANK YOU



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